



NATIONAL UNIVERSITY

of Computer & Emerging Sciences

FYP Mid -1 Report Document

“F24-010-D-TalkTracks”

(Development)



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Introduction

Problem Overview

Problem Statement and Motivation

In this fast-paced world, time-efficient tech solutions are necessary to reduce human effort where having efficient tools has become a necessity. Tools to process text have been around for quite some time for syntactic and semantic search in text, text-summarization, text-paraphrase, etc., such as ‘*QuillBot*’ (QuillBot, n.d.). However, targeting audio and video files is rare.

Tools to summarize, generate subtitles, search through audio and video files are rare and sometimes inefficient. For example, one of the famous sites, ‘*PicoVoice*’ (PicoVoice, n.d.), lacks efficient search in fast-paced audios. This leads to enormous problems since audio and video files have become a common part of our lives, whether it be voice notes, video lectures, podcasts or news, all these files are used in our daily-lives and are required to be efficiently processed to facilitate easy use.

Search through audio/video files have become significant due the common use of these files. For example, if a student needs to quickly search through a video lecture to find a key word or maybe they are looking for a similar concept but are not aware of the exact words used in the file, they require syntactic or semantic search through this file. Along with these demands, concern for audio/video summarization arises too.

Coherent Summarization of the text generated from an audio is one of the most challenging tasks for acoustic engineers nowadays. This is due to the increasing complexity of audio processing having divergent chat dialogues, comprehensive educational lecture details or extensive discussions on hot topics with a wide variety of vocabulary involved. Especially when it comes to narrowing down the major highlights from an audio for extracting the right chunk of information, many existing tools provide extractive summarization, whereas the desired outcome is context-rich abstractive summarization.

Problem Solution

Our strategic solution to these challenges is “Talk Tracks”. A tool that provides **syntactic** as well as **semantic search** through audio/video lecture files, for accelerated lecture understanding with just a file upload. Such searches allow users to search a definition from a

lecture, a quote or maybe a reference whether they are acquainted with the precise wording or not.

Talk Tracks provides users with the most precise **extractive summarization** of the input audio/video file when they desire an exact portion from the file. Our web application also provides **abstractive summarization** option to enable the users to swiftly skim through the files. Such an abstractive summary becomes crucial when users desire to get the overview of the lecture. This delivers a concise overview of the file to the users, conserving their time and energy.

Along with these perks, Talk Tracks also enables hearing-impaired people with to easily read the text generated from audio lecture, read the summary of the lecture or even interact with the audio through a **chatbot**, a chatbot that allow users to ask queries related to the audio. (Patil, Patil, Aware, & Purushartha)

In addition to these solutions, our web application also provides **subtitle generation** and **noise reduction** to gather all features upon a single platform for user-convenience.

Project Scope

Users will have to navigate to the main page of Talk Tracks and access features of our web app. On the Home Page, the user will be free to explore all features such as Search through audio/video files, Generate Summary, Generate Subtitles and Reduce Noise, purchase packages or manage their profile.

If the user desires to explore or access a feature, they will click on the button for that feature and will be redirected to the feature page. On this page, the user will upload an audio/video file or even record their own audio at the spot. Now let us observe what the further steps would be for each feature.

For example, the user desires to Search through audio/video files, after uploading the file, the user will enter the upload button, then a search field will appear for the user to enter a word, phrase or record their voice. The user will have the option to search this input syntactically or semantically. Wherever the matched text (semantic or syntactic) lies, those audio/video timestamps will get highlighted along with the matched phrase. In addition, a chatbot will be available for the user to interact with their file and extract information efficiently from the file. User will ask queries to the bot and receive answers present in the file.

If the user desires to Generate summary of the audio/video file, after uploading the file, the user will have the option to generate extractive or abstractive summarization, once it is selected, summary will be generated. The user will download this summary by clicking the download button. Similar steps will be taken for subtitle generation and noise reduction as well.

All the files uploaded, and the processing done on the files will be stored on the user profile and the user can manage their profiles by creating their accounts.

Modules

We have five modules in our project.

1. Semantic Search:

This module allows user to upload audio or video file and search related or similar text through the file. This makes it incredibly time-saving and convenient for the users to identify the portion of the input they wish to detect or acquire.

2. Syntactic Search:

This module allows user to upload audio or video file and search exact text through the file. This allows user to quickly locate specific information within the multimedia provided.

3. Summary Generation:

This module provides the option of generating extractive as well as abstractive summary of any audio or video related to contexts such as lectures, day to day conversations, podcasts, poems, ...

4. Noise Reduction:

This module refers to all the noise reduction algorithms applied to the provided input by the user to remove unwanted background noises increasing the quality of the audio while creating a more comfortable listening environment.

5. Subtitle Generation:

This module refers to the APIs used for subtitle generation for the provided audio or video with time stamps.

6. Chatbot Assistant:

This module focuses on deploying an intelligent chatbot that provides an interactive user experience by responding to queries related to the user input using different APIs to gather relevant information.

7. User Profile Management:

This module ensures to maintain user profile, the account made by the user. Files uploaded by the user are maintained in the user profile along with the processing done on them. In the profile, user can maintain their packages as well.

8. Admin Portal Management:

This module enables admin to maintain their own as well as users' profiles, subscription packages and view reports, daily report, weekly report, or annual report and

9. Payment Management:

This module enables user to pay for packages. Talk Tracks offers three packages for users: Starter, Pro and Enterprise. User can purchase any package and view the payment details in their profile.

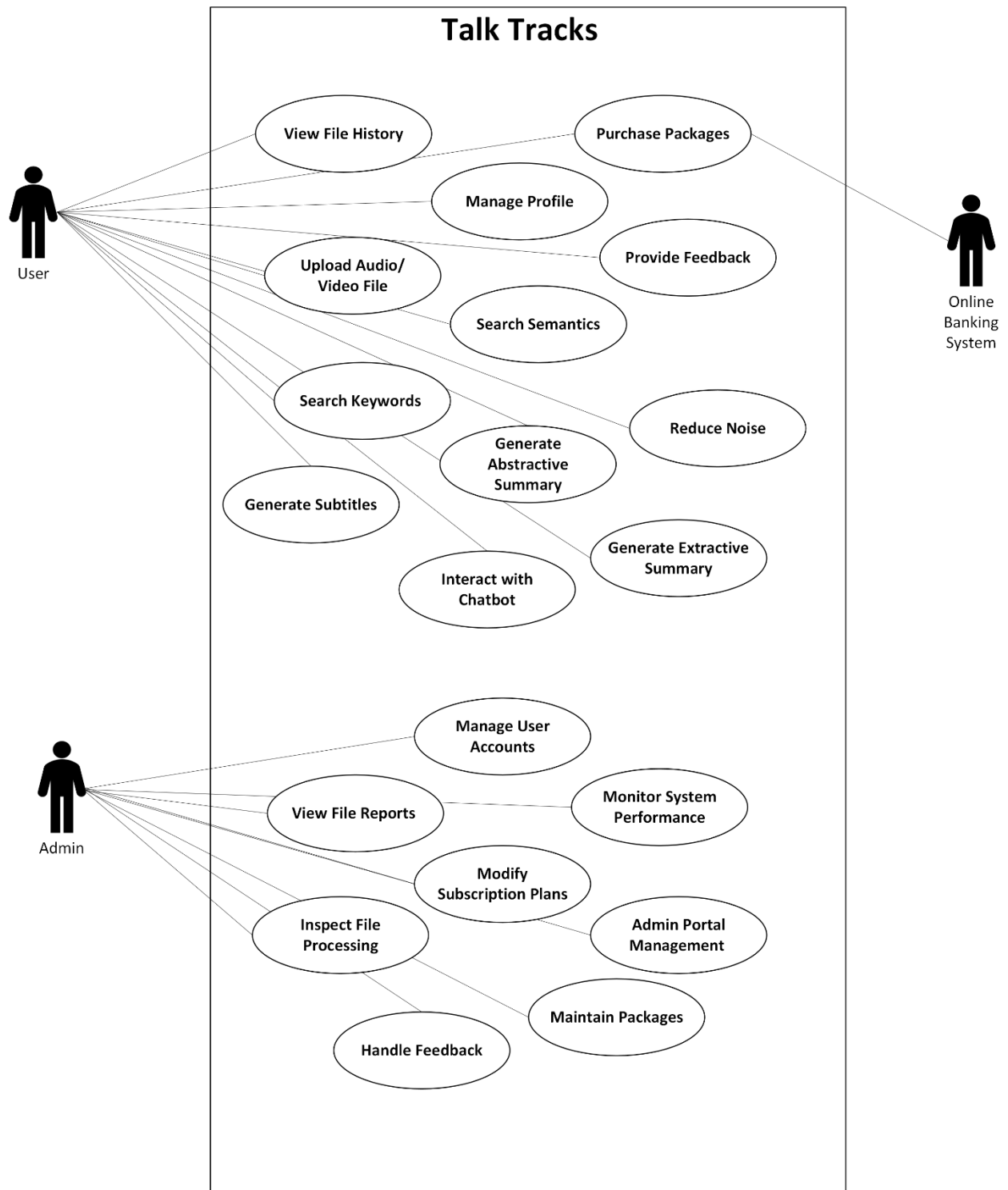
User Classes and Characteristics

User Classes	Description
Talk Tracks	Talk Tracks is the controller class of the project which contains all the features that the project offers such as summary generation, subtitle generation, syntactic and semantic search, chatbot assistant and noise reduction. Talk Tracks offers different packages for the end users to avail. End users interact with Talk Tracks to perform processing and admin manages Talk Tracks.
Admin	Admin oversees the system manages packages that the app offers as well as the end user details and payment details. Admin also monitors user feedback.

End User	End user interacts with the system to upload audio or file and apply different processing on the uploaded file.
Package	Packages are the various subscription plans that the application offers its users to process different number of files.
Payment	Payment class handles the billing from the end user for a package.
Feature	Feature is the class that parents all the processing classes like noise reducer, summarizer, searcher, subtitle generator and chatbot assistant.
Subtitle Generator	Subtitle generator generates accurate subtitles for the audio or video file uploaded by the end user.
Noise Reducer	Noise reduces the noise of the audio/video file uploaded by the end user.
Searcher	Searcher allows the end user to search semantically as well as syntactically in the provided audio/video file.
Summarizer	Summarizer provides abstractive as well as extractive summary of the audio/video file.
Chatbot	Chatbot assistant allows the user to ask relevant questions about the file uploaded by the end user.
File	File class represent the audio/video file that the end user uploads for the processing.
Pre-processed File	Pre-processed file converts the audio/video file to text for further application of the provided features by the application.

Project Requirements

Use Case Diagram



Use Cases (Iteration 1)

1. Syntactic Search

High Level/Brief Use Case:

Use Case	Search Keywords
Actor	User
Type	Primary
Description	User uploads audio/video file on the webpage. The file is processed, user enters the keyword to search within the file and the relevant keyword timestamps are highlighted. These uploaded and processed files are saved on the user's profile for later use.

Fully Dressed Use Case

- i. **Use Case:** Search Keywords
- ii. **Scope:** Talk Tracks Web Application
- iii. **Level:** User Goal
- iv. **Primary Actor:** End-User
- v. **Stakeholders & Interests:**
 - End-User: wants to be able to search desired keywords from the uploaded audio/video lecture file.
 - Admin: wants to provide effective search from the file and make any amendments in the feature if required.
- vi. **Pre-Conditions:**
 - End-user should be logged in.
 - End-user should have a package selected and purchased accordingly.
- vii. **Post-Conditions (Success Guarantee):**
 - The uploaded and processed files should be saved in end-user's profile.
- viii. **Main Success Scenario:**

End-User Action

Talk Tracks' Responsibility

1. User arrives at Syntactic Search Feature web page.
2. System displays upload file/record option to the user.
3. User uploads audio/video file or records audio to search keywords within.
4. System processes the file.
5. User enters keyword to search within the file or voice records the keyword.
6. User searches for the input keyword.
7. System highlights the searched keyword instances w.r.t the audio/video timestamps.
8. System saves the uploaded and processed file in the user's profile.
9. User leaves the website.

ix. Extensions:

- If the user mistakenly uploads incorrect file, they can remove the chosen one and reselect the desired files.
- In case of connection loss, the user shall reconnect to internet and begin from where they left off.
- If the file is not uploaded due to lack of connection, user would have to reupload the file.
- If no match for the keyword is found, the system displays “Match not found”.

x. Frequency: multiple time upon an uploaded lecture file.

2. Noise Reduction

High Level/Brief Use Case

Use Case	Reduce Noise
Actor	User
Type	Primary

Description	User uploads audio/video file on the webpage. The file is processed and noise present in the file is reduced. This processed file can be downloaded by the user and is saved in their profile as well.
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Fully Dressed Use Case

- i. **Use Case:** Reduce Noise
- ii. **Scope:** Talk Tracks Web Application
- iii. **Level:** User Goal
- iv. **Primary Actor:** End-User
- v. **Stakeholders & Interests:**
 - End-User: wants to be able to reduce noise from the uploaded audio/video lecture file.
 - Admin: wants to provide effective noise reduction of the file and make any amendments in the feature if required.
- vi. **Pre-Conditions:**
 - End-user should be logged in.
 - End-user should have a package selected and purchased accordingly.
- vii. **Post-Conditions (Success Guarantee):**
 - The uploaded and processed files should be saved in end-user's profile.
- viii. **Main Success Scenario:**

End-User Action	Talk Tracks' Responsibility
1. User arrives at Noise Reduction Feature web page.	2. System displays upload file/record option to the user.
3. User uploads audio/video file or records audio to reduce noise from.	4. System processes the file.
	5. System displays the file with reduced noise.
6. User can listen or watch the processed file.	

7. User can download this file.

8. System saves the uploaded and processed file in the user's profile.

9. User leaves the website.

ix. Extensions:

- If the user mistakenly uploads incorrect file, they can remove the chosen one and reselect the desired files.
- In case of connection loss, the user shall reconnect to internet and begin from where they left off.
- If the file is not uploaded due to lack of connection, user would have to reupload the file.
- If no match for the keyword is found, the system displays "Match not found".

x. Frequency: single time upon an uploaded lecture file.

Functional Requirements (FRs)

1. (Module) Semantic Search

- The system shall provide user to upload audio files (e.g., MP3, WAV) or video files (e.g., MP4).
- The system should be able to accurately transcribe audio to text.
- The system should allow users to enter text queries for semantically searching within audio/video content.
- The user should highlight the timestamps in the multimedia file where the text is present.
- The system should allow user to generate report and download it as a pdf.

2. (Module) Syntactic Search

- The system shall provide user to upload audio files (e.g., MP3, WAV) or video files (e.g., MP4).
- The system should be able to accurately transcribe audio to text.
- The system should allow users to enter text queries for syntactically searching within audio/video content.

- The user should highlight the timestamps in the multimedia file where the text is present.
- The system should allow user to generate report and download it as a pdf.

3. (Module) Summary Generation

- The system shall provide user to upload the file (audio or video).
- The system should be able to accurately transcribe audio to text.
- The system should allow user to download the generated summary as pdf.

4. (Module) Noise Reduction

- The system shall provide user to upload the file (audio or video).
- The system should allow user to download the processed audio in the form of mp3.

5. (Module) Subtitle Generation

- The system shall provide user to upload the file (audio or video).
- The system should be able to accurately transcribe audio to text.
- The system should allow user to download the generated subtitles as .srt.

6. (Module) Chatbot Assistant

- The system should allow the user to ask queries related to the multimedia input provided.

Non-Functional Requirements (NFRs)

The non-functional requirements for Talk Tracks are as follow.

1. Usability Requirements

USE-1: Users shall be able to retrieve accurate results from the processed audios/videos.

2. Performance Requirements

PERF-1: The system shall process a video of 3 minutes in 3 minutes.

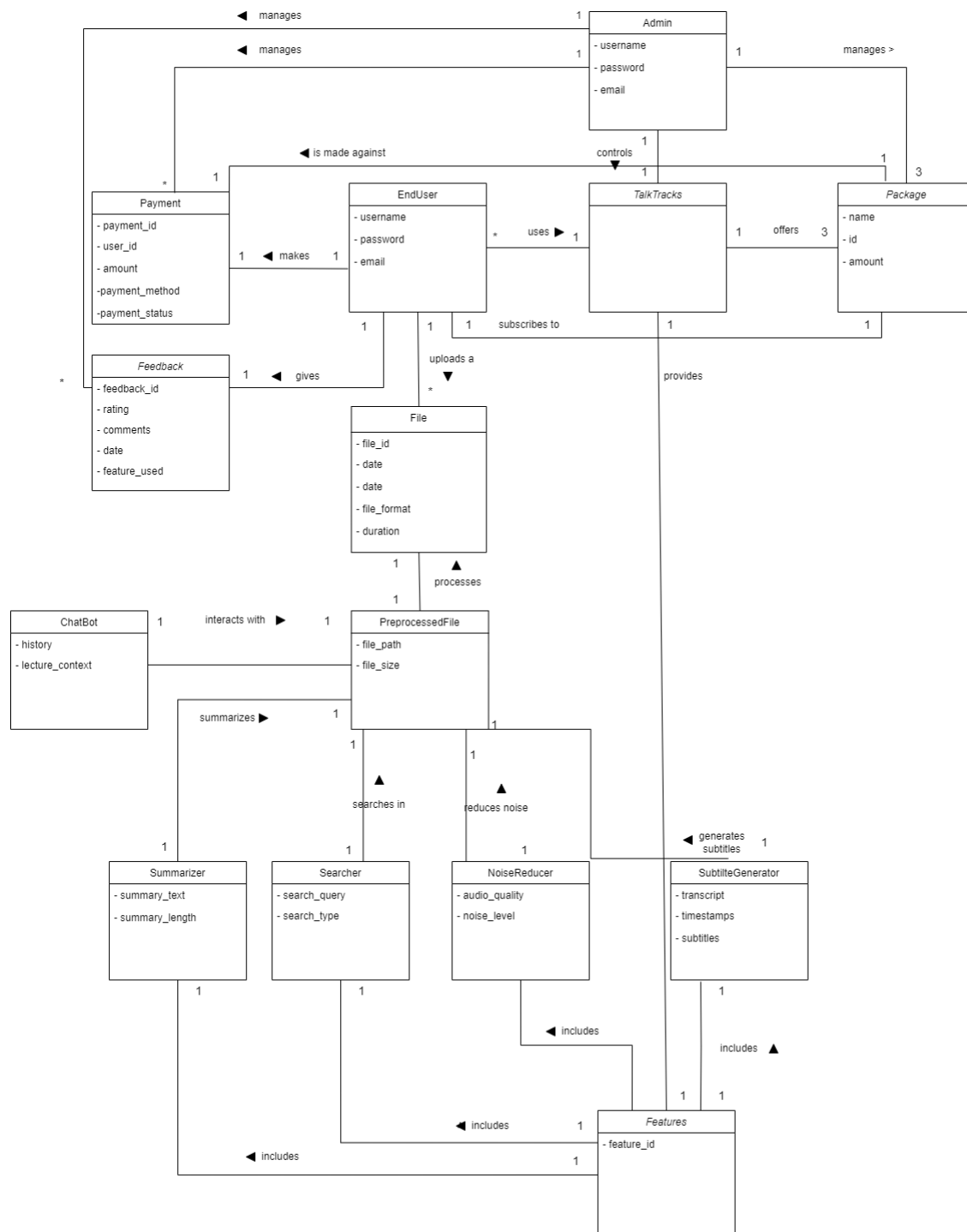
3. Compatibility Requirements

COM-1: The system shall be compatible with audio formats such as MP3 and WAV and video formats such as MP4 and MWAV.

4. Security Requirements

SEC-1: Protecting data and granting access only to authorized users by verifying their identity through 2-Factor Authentication.

Domain Model



System Overview

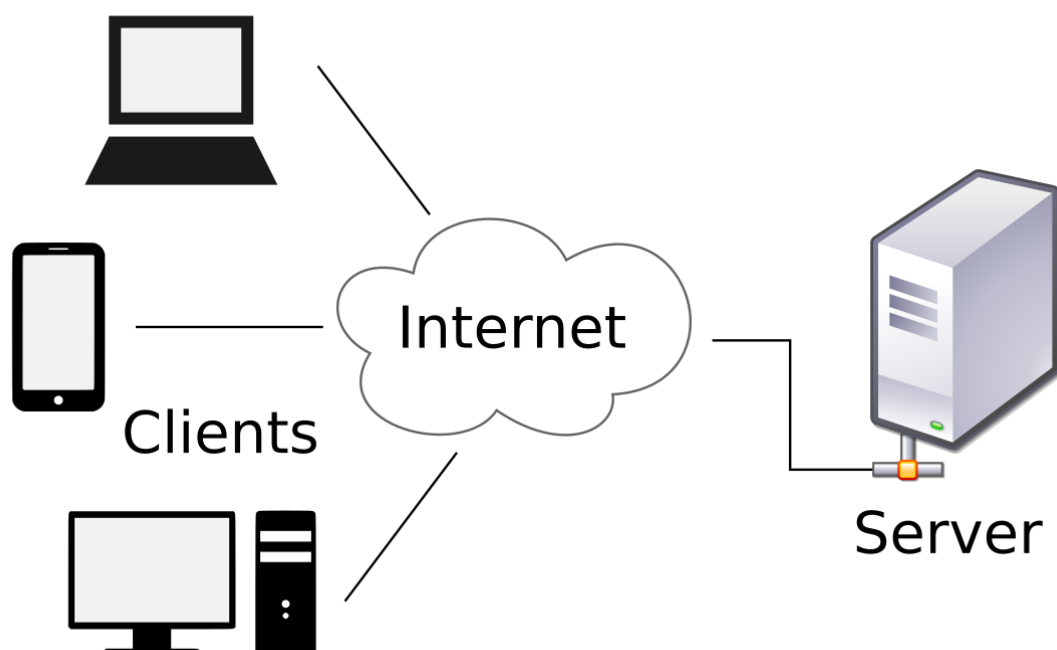
Architectural Design

The chosen software architecture for Talk Tracks is **Client-Server Architecture**. The reason for choice and diagram for the architecture is listed below.

Reason

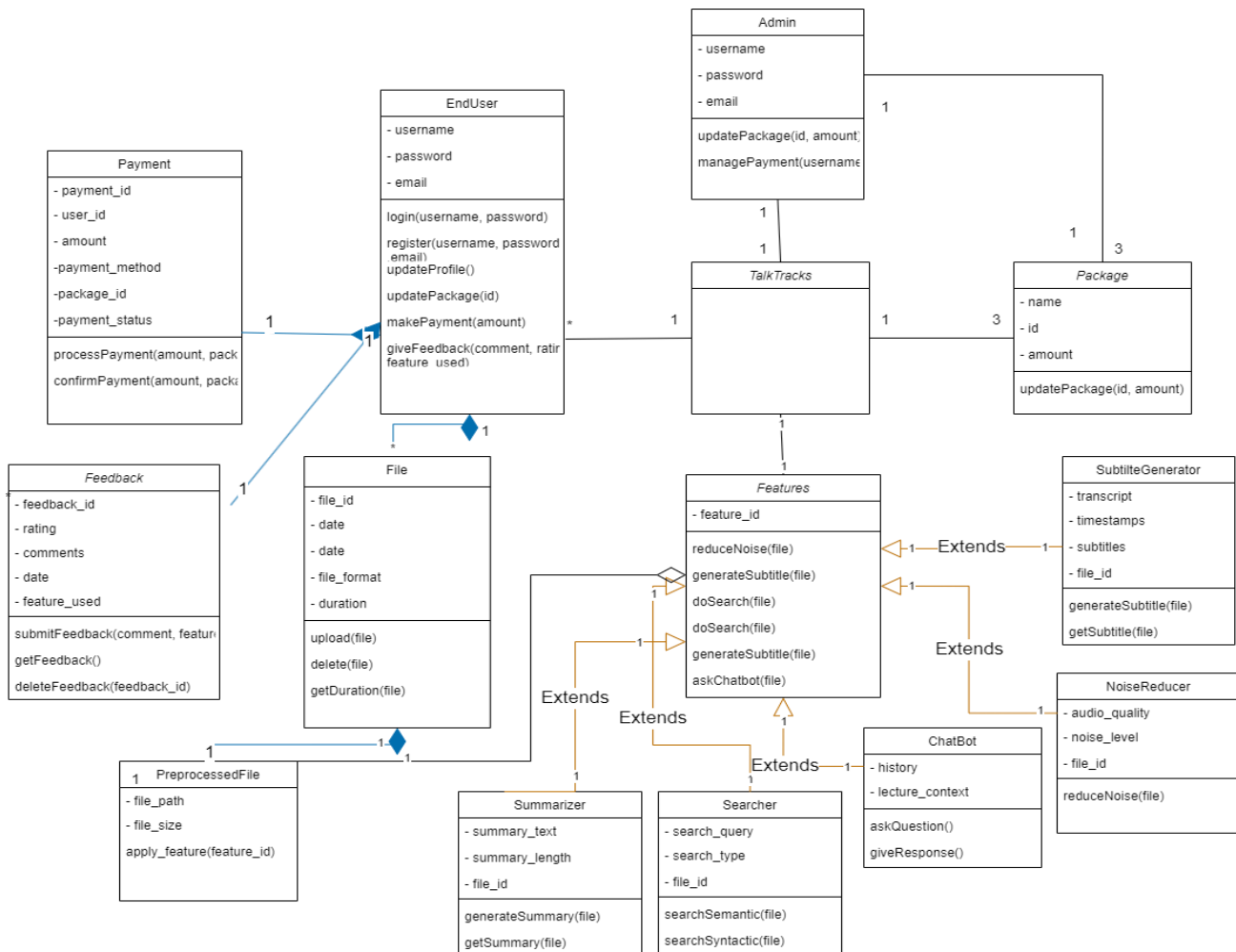
Client-Server Architecture is chosen because in the Talk Tracks web app the end-user uploads files using the user interface (front-end), the files are processed at the back end and stored at the server, resulting in client-server architecture.

Diagram



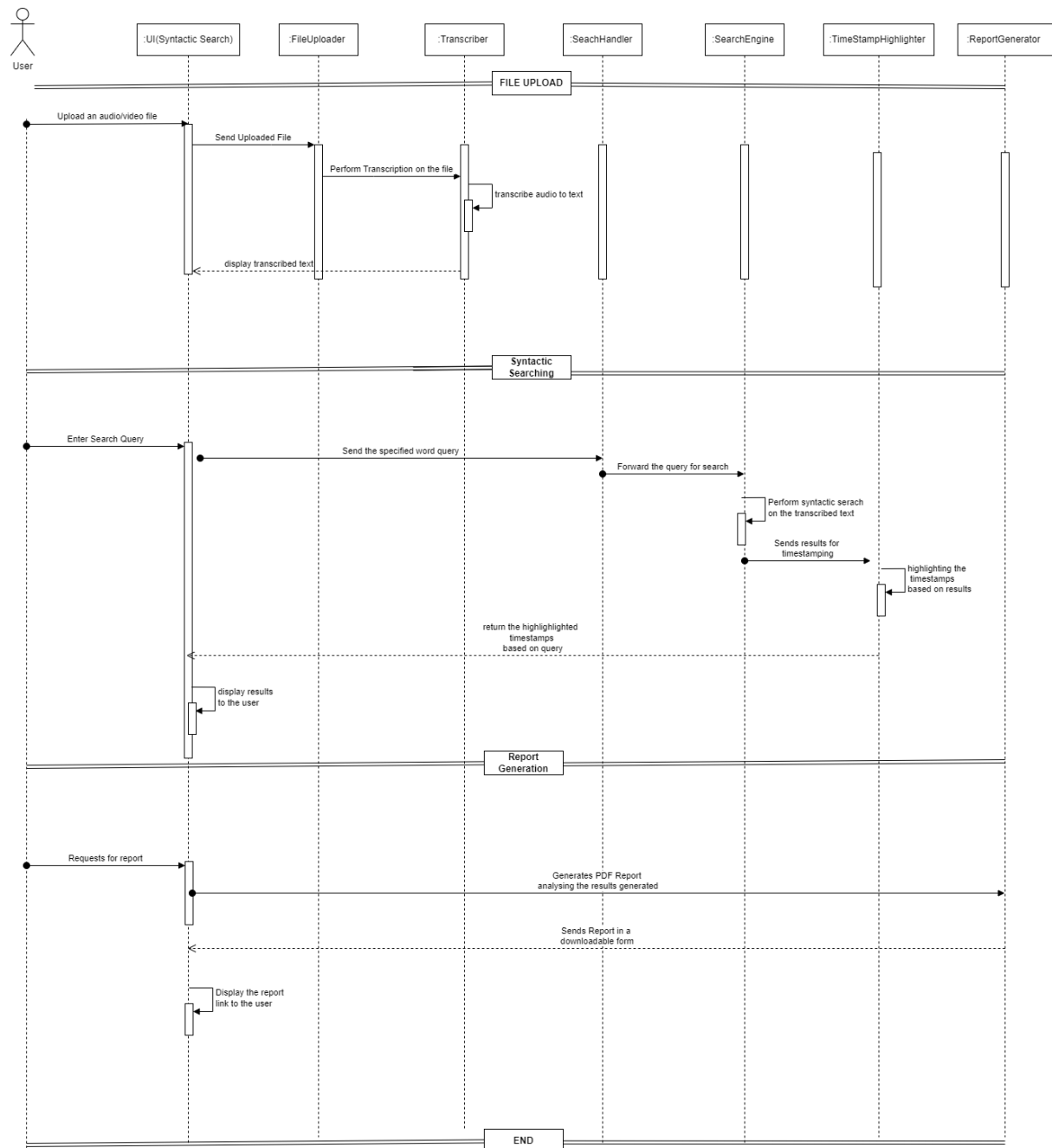
Design Models

Class Diagram

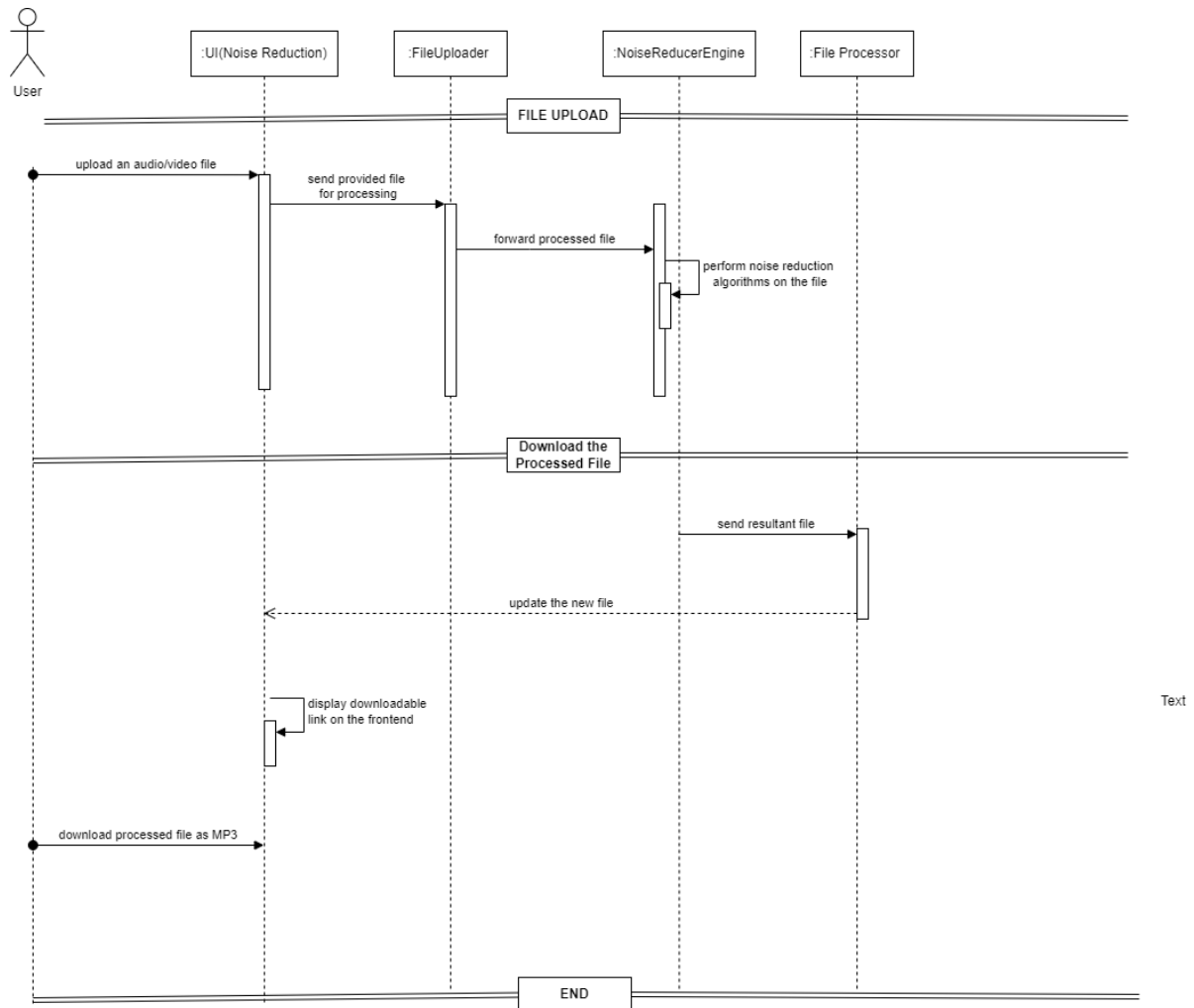


Sequence Diagrams (Iteration 1)

1. Syntactic Search

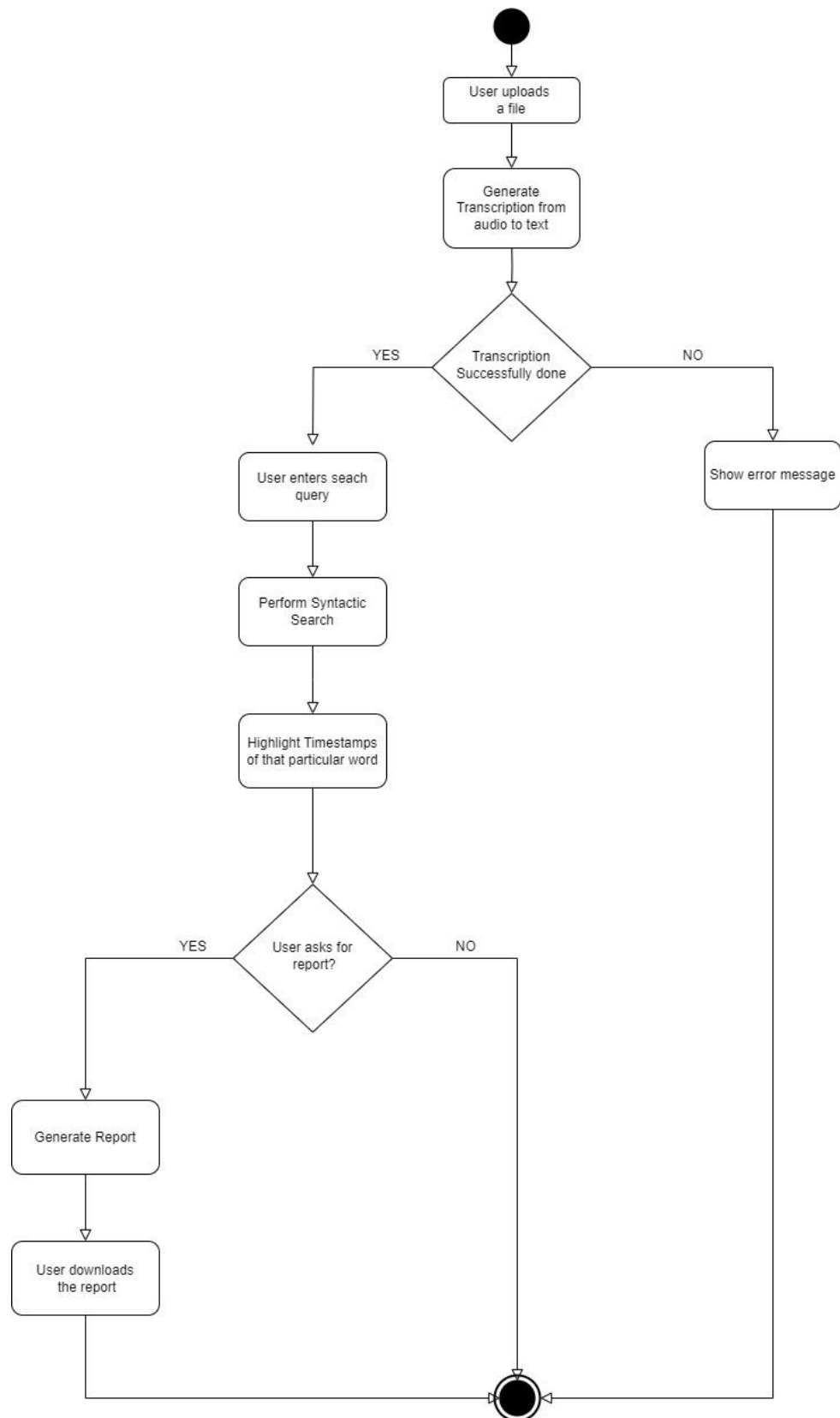


2. Noise Reduction

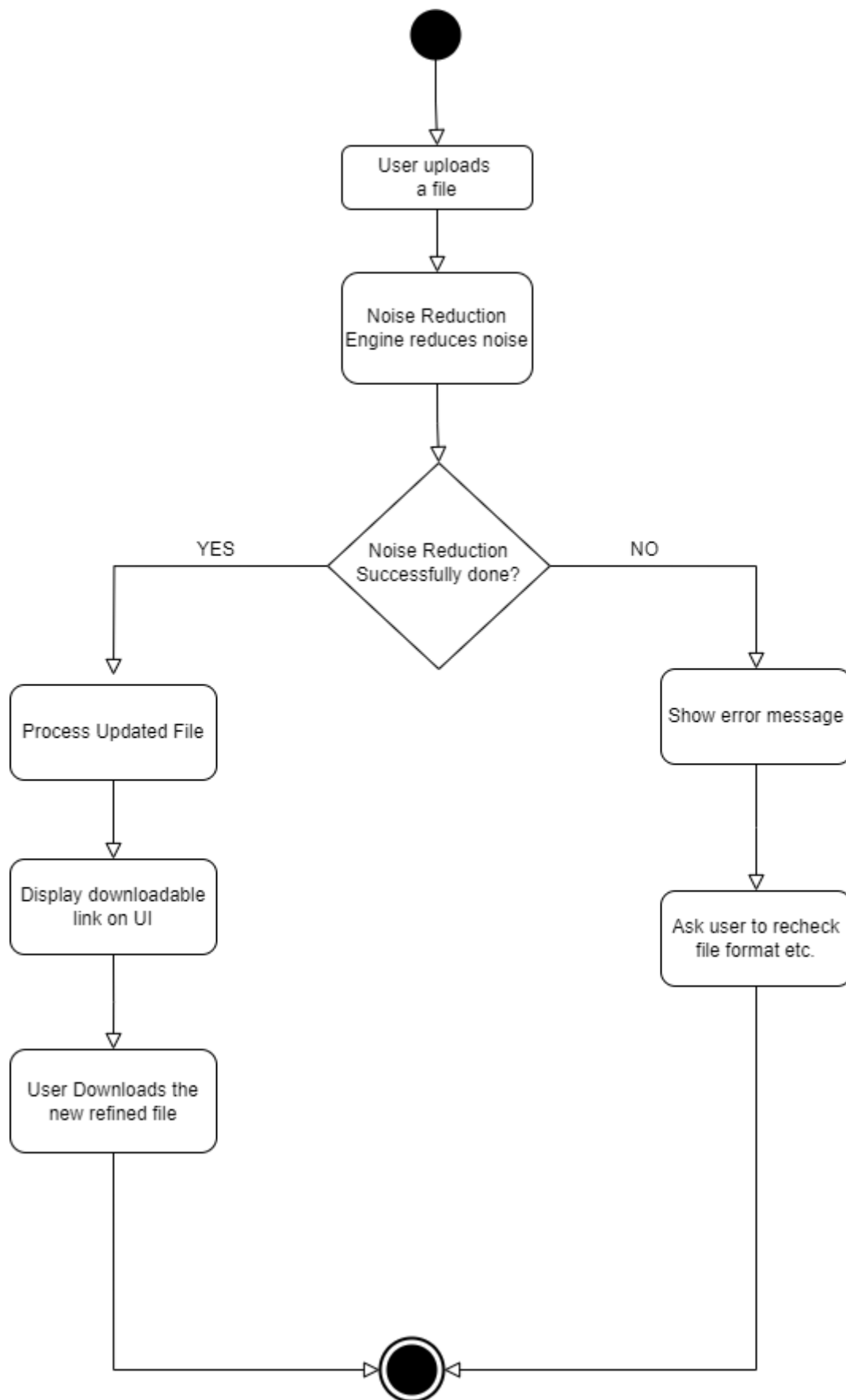


Activity Diagrams (Iteration 1)

1. Syntactic Search

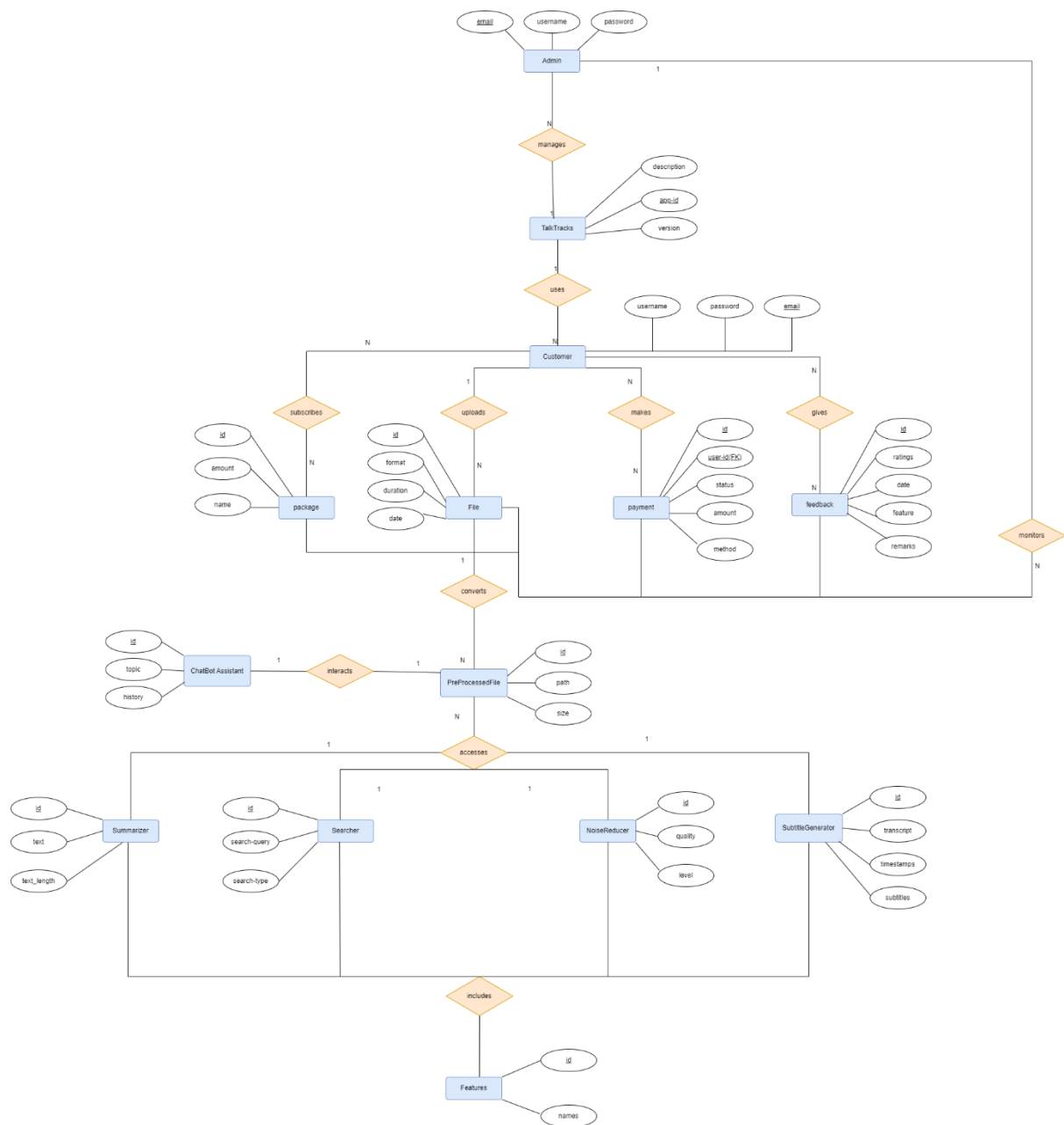


2. Noise Reduction



Data Design

ERD



Iteration Achievement

1. Syntactic Search

- Algo used – Boyer Moore Algorithm.
- User interface for the feature is implemented.
 - User uploads file and the file is processed.
 - User can search for the keyword.

- The processing on the audio/video is done file using python and the output is directed to our react application using the 'Flask' framework.
- Tools & Technologies used:
 - React
 - Python – Flask
 - Bootstrap

2. Noise Reduction

- User interface for the feature is implemented.
- User uploads file and the file is processed.
- Tools & Technologies used:
 - Using 'noisereuce' and 'scipy.io.wavfile' from python, audio files are processed with cleaner and understandable audio.
 - React
 - Python – Flask
 - Bootstrap

3. UI/UX

- Login / Signup
 - User interface made.
 - Database connected and the credentials successfully stored.
 - Input validation ensured.
 - Security mechanism – hashing applied.
 - Hashing generates an encrypted token known only to the server.
- Tools & Technologies used:
 - React
 - Bootstrap
 - Express Js
 - MongoDB Atlas
- All pages for the web app created and routed.
- Basic prototype for the end user implemented.

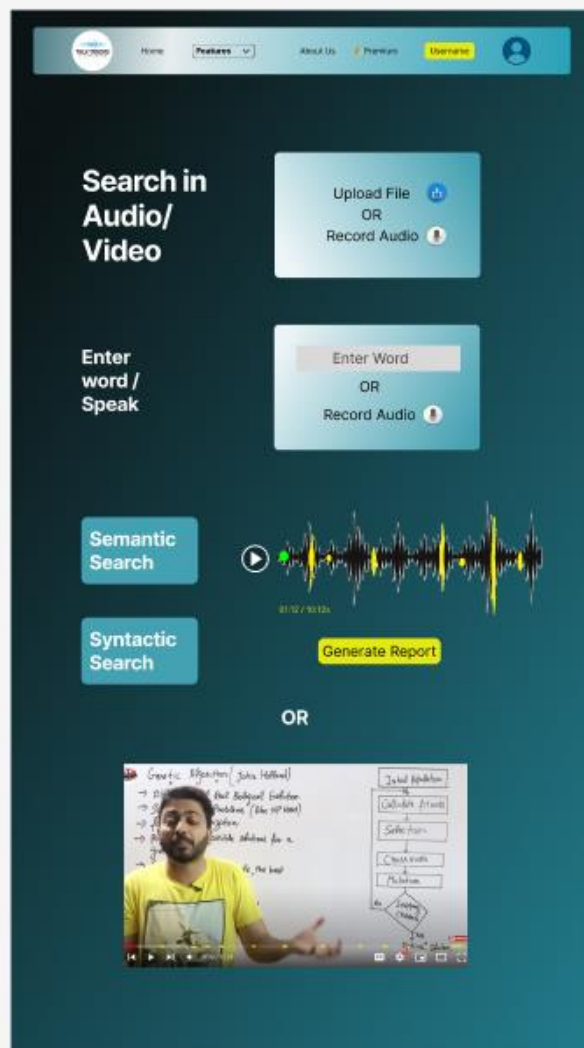
4. Prototype

Below are the attached prototyped web pages created for Talk Tracks web application created on *Figma* (Figma, n.d.). All the prototypes are made via frame.

End User interface:



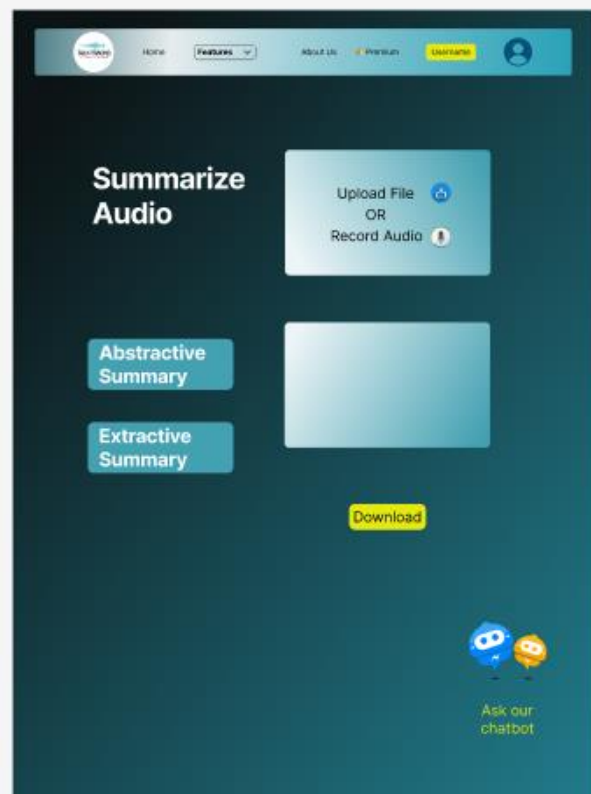
Search



The Search interface features a dark teal background with a navigation bar at the top containing links for Home, Features, About Us, Premium, and a User profile icon. The main content area is divided into several sections:

- Search in Audio/Video**: A section with two options: "Upload File" (with a file icon) or "Record Audio" (with a microphone icon).
- Enter word / Speak**: A section with two options: "Enter Word" (with a text input field) or "Record Audio" (with a microphone icon).
- Semantic Search**: A section with a play button icon and a waveform visualization.
- Syntactic Search**: A section with a "Generate Report" button.
- OR**: A central separator.
- Video Player**: A video player showing a man in a yellow shirt speaking, with a whiteboard in the background.

Summarize



The Summarize interface features a dark teal background with a navigation bar at the top containing links for Home, Features, About Us, Premium, and a User profile icon. The main content area is divided into several sections:

- Summarize Audio**: A section with two options: "Upload File" (with a file icon) or "Record Audio" (with a microphone icon).
- Abstractive Summary**: A section with a large empty box for the summary.
- Extractive Summary**: A section with a large empty box for the summary.
- Download**: A yellow button.
- Ask our chatbot**: A section with a chatbot icon and the text "Ask our chatbot".

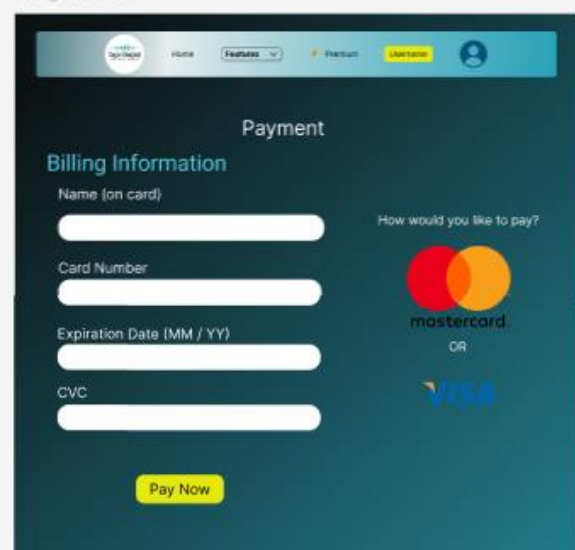
Membership



The Membership interface features a dark teal background with a navigation bar at the top containing links for Home, Features, About Us, Premium, and a User profile icon. The main content area displays three membership plans:

US\$15.99/mo	US\$20.99/mo	US\$50.99/mo
process 20 audios per week	process 50 audios per week	process unlimited audios
Buy	Buy	Buy

Payment



The Payment interface features a dark teal background with a navigation bar at the top containing links for Home, Features, About Us, Premium, and a User profile icon. The main content area is titled "Payment" and includes a "Billing Information" section:

Billing Information

Name (on card)

Card Number

Expiration Date (MM / YY)

CVC

[Pay Now](#)

How would you like to pay?

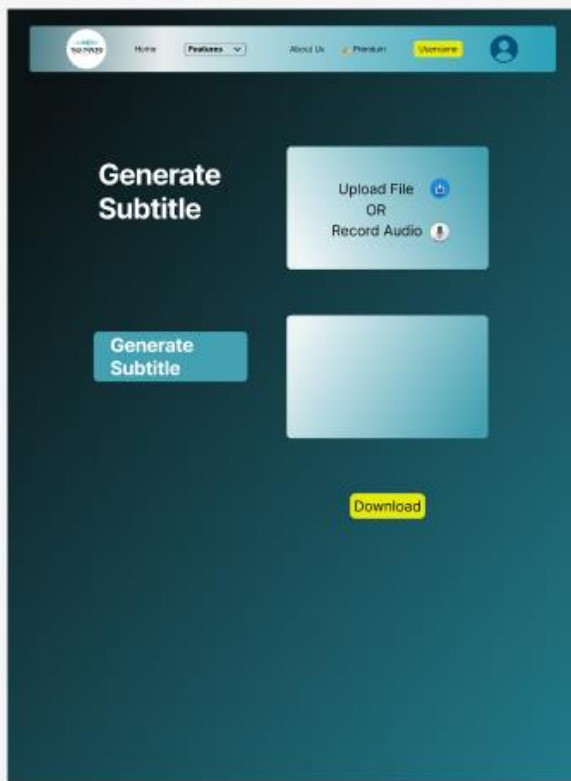
 **mastercard**

OR

 **visa**

Afterlogin

Generate subtitle



Phonic Mavens Home Features About Us Premium Downloads

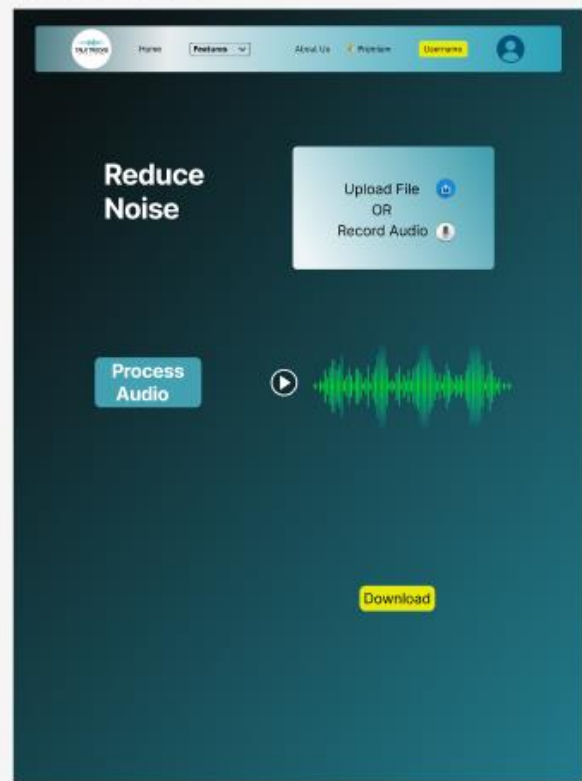
Generate Subtitle

Upload File  OR Record Audio 

Generate Subtitle

Download

Reduce Noise



Phonic Mavens Home Features About Us Premium Downloads

Reduce Noise

Upload File  OR Record Audio 

Process Audio  

Download

Profile



Phonic Mavens Home Features About Us Premium Downloads

User Profile

Profile

Plan Purchased

Extra Features Avail



Username: Phonic Mavens
Email: phonic.mavens@gmail.com
Password:

Edit Profile

Subtitles



Phonic Mavens Home Features About Us Premium Downloads

Subtitles Generated

Files



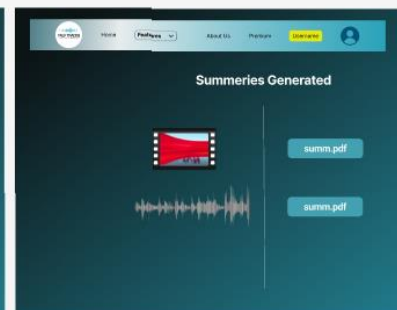


Processed

sub.art

sub.art

Summaries



Phonic Mavens Home Features About Us Premium Downloads

Summaries Generated

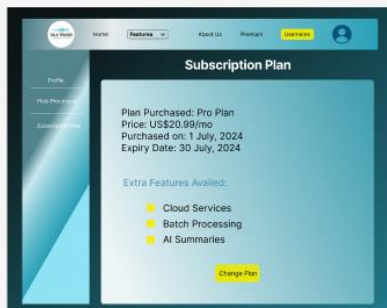




summ.pdf

summ.pdf

Subscription Plan



Phonic Mavens Home Features About Us Premium Downloads

Subscription Plan

Profile

Plan Purchased

Extra Features Avail

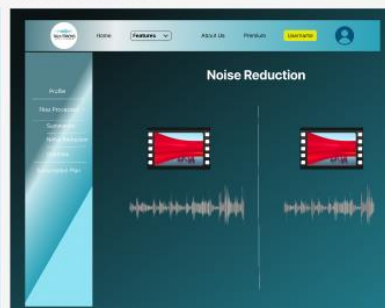
Plan Purchased: Pro Plan
Price: US\$20.99/mo
Purchased on: 1 July, 2024
Expiry Date: 30 July, 2024

Extra Features Avail:

- Cloud Services
- Batch Processing
- AI Summaries

Change Plan

Noise Reduction



Phonic Mavens Home Features About Us Premium Downloads

Noise Reduction

Profile

Plan Purchased

Extra Features Avail



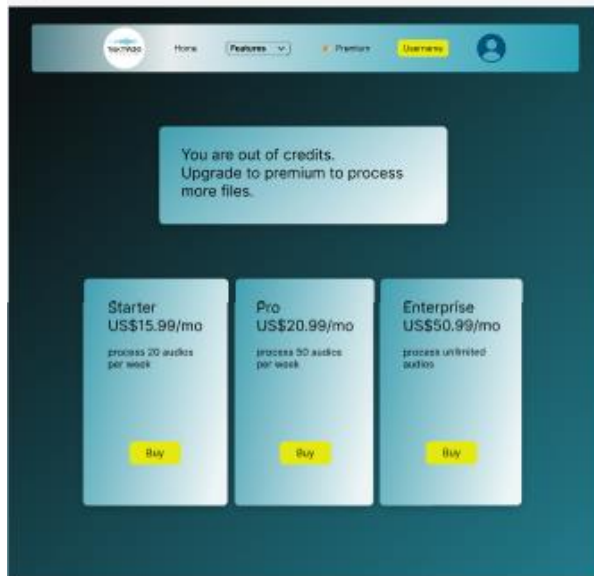






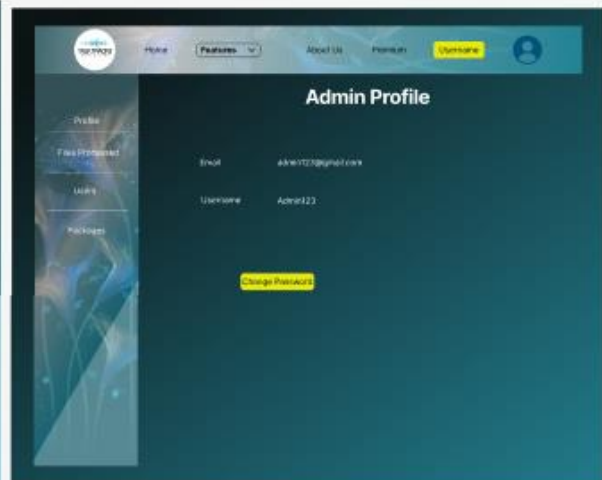
Admin Portal interface:

Expired Credits



The 'Expired Credits' screen features a dark teal background. At the top, a navigation bar includes a logo, 'Home', 'Features', 'Premium', and 'Username' links. A central white box contains the message: 'You are out of credits. Upgrade to premium to process more files.' Below this, three pricing cards are displayed: 'Starter' at US\$15.99/mo (20 audios/week), 'Pro' at US\$20.99/mo (50 audios/week), and 'Enterprise' at US\$50.99/mo (unlimited audios). Each card has a yellow 'Buy' button.

Admin



The 'Admin' screen shows an 'Admin Profile' section. It includes a sidebar with 'Profile', 'Files Processed', 'Users', and 'Packages'. The main content area displays the email 'admin123@gmail.com' and the username 'Admin123', with a yellow 'Change Password' button below.

Admin-files weekly

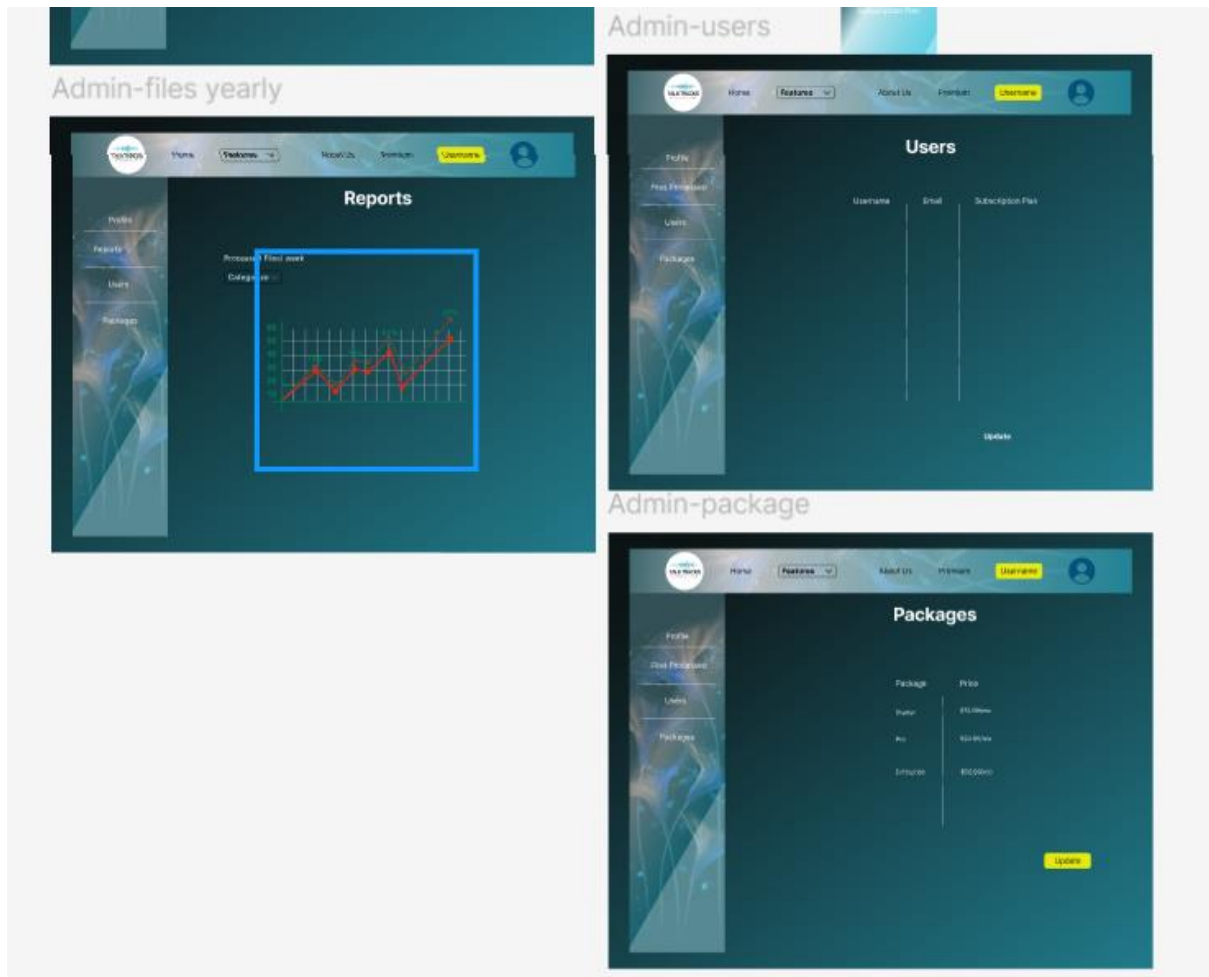


The 'Admin-files weekly' screen displays a 'Reports' section. It features a sidebar with 'Profile', 'Reports', 'Users', and 'Packages'. The main content area shows 'Processed Files/week' and a 'Categories' dropdown. A line graph with red data points and green trend lines is visible.

Admin-files monthly



The 'Admin-files monthly' screen displays a 'Reports' section, similar to the weekly view. It includes a sidebar with 'Profile', 'Reports', 'Users', and 'Packages'. The main content area shows 'Processed Files/month' and a 'Categories' dropdown. A line graph with red data points and green trend lines is visible.



References

Figma. (n.d.). Retrieved from <https://www.figma.com/>.

Patil, S., Patil, S., Aware, P., & Purusharthi, Y. (n.d.). PODCAST ACCESSIBILITY THROUGH TRANSCRIPTION AND SUMMARIZATION WITH SPEECH SYNTHESIS. 06/Issue:02/February-2024. Retrieved from https://www.irjmets.com/uploadedfiles/paper//issue_2_february_2024/49230/final/fin_irjmets1707589470.pdf

PicoVoice. (n.d.). Retrieved from <https://picovoice.ai/blog/voice-search/>

QuillBot. (n.d.). Retrieved from <https://quillbot.com/summarize>